These activities show that certain things are necessary for photosynthesis. They are:

1. Chlorophyll - Green pigment in leaves
2. Water
3. Carbon dioxide (from air)
4. Sun light

19.4 Transpiration

The loss of water in the form of water vapour from the aerial parts of the plant body is called as transpiration. The leaves have tiny, microscopic pores called **stomata**. Water evaporates through these stomata. Each stomata is surrounded by guard cells. These guard cells help in regulating the rate of transpiration by opening and closing of stomata.

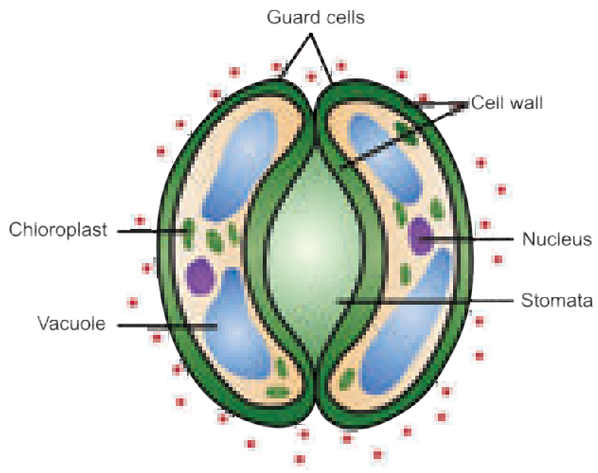


Figure 19.6 Structure of Stomata

19.4.1 Types of Transpiration

There are three types of transpiration:

Stomatal transpiration: Loss of water from plants through stomata. It accounts for 90-95% of the water transpired from leaves.

Cuticular transpiration: Loss of water in plants through the cuticle.

Lenticular transpiration: Loss of water from plants as vapour through the lenticels. The lenticels are tiny openings that protrude from the barks in woody stems and twigs as well as in other plant organs.

But transpiration is necessary for the following reasons.

1. It creates a pull in leaf and stem.
2. It creates an absorption force in roots.
3. It is necessary for continuous supply of minerals.
4. It regulates the temperature of the plant.

Activity 5

Take a plastic bag and tie it over a leaf and place the plant in light. You can see water condensing inside the plastic bag. The water is let out by the leaves. Why does this occur?



19.4.2 Exchange of Gases

How does the plant get air? The leaves have minute pores called **stomata** through which the exchange of air takes place. These minute pores can be seen through a microscope. Air exchange takes place continuously through the stomata. Plants exchange gases (CO_2 to O_2) continuously through these stomata. You will study more about these physiological process in your higher classes.

Points to Remember

- Growth movement whose direction is determined by the direction of the stimulus is called Tropism.
- Non-directional, response of a plant part to stimulus is called nastic movement.
- The process by which plants prepare their food material is called photosynthesis.
- The loss of water in the form of water vapour from the aerial parts of the plant body is called transpiration.
- Stomata are minute opening on the leaves.

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